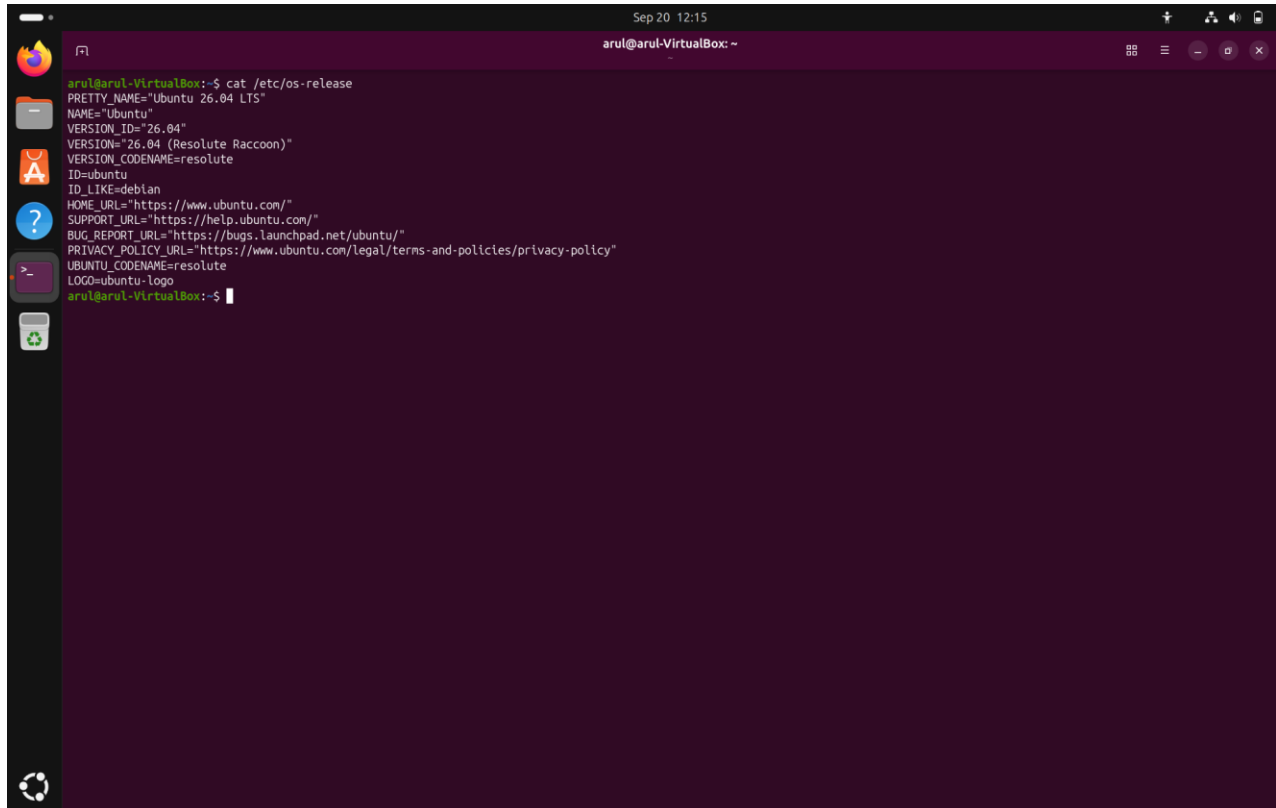


Devops -Week-1 Assignment

1. Run a command to check the Linux distribution name and version of the operating system installed on your machine



The screenshot shows a terminal window titled 'arul@arul-VirtualBox: ~' with a timestamp of 'Sep 20 12:15'. The user has executed the command 'cat /etc/os-release'. The output displays the following information:

```
arul@arul-VirtualBox:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
arul@arul-VirtualBox:~$
```

- Cat- view file content
- /etc- system configuration files
- os- release - operating system

Files & Directories:

- Create directory /home/devops project
- Inside it, create 2 files: plan.txt and network_report.txt
- Copy plan.txt to /tmp and delete the original

```
arul@arul-VirtualBox:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
arul@arul-VirtualBox:~$ pwd
/home/arul
arul@arul-VirtualBox:~$ mkdir devops_project
arul@arul-VirtualBox:~$ ls
Desktop Documents Downloads ForumWall Music Pictures ProzoneWall Public Templates Videos devops_project enc enc.txt enc2 snap
arul@arul-VirtualBox:~$ ls -ld
drwxr-x--- 21 arul arul 4096 Sep 20 12:20 .
arul@arul-VirtualBox:~$ cd devops_project
arul@arul-VirtualBox:~/devops_project$ touch plan.txt
arul@arul-VirtualBox:~/devops_project$ ls
plan.txt
arul@arul-VirtualBox:~/devops_project$ touch network_report.txt
arul@arul-VirtualBox:~/devops_project$ ls
network_report.txt plan.txt
arul@arul-VirtualBox:~/devops_project$ pwd
/home/arul/devops_project
arul@arul-VirtualBox:~/devops_project$ cp /home/arul/devops_project/plan.txt /tmp/
arul@arul-VirtualBox:~/devops_project$ ls -l /tmp/plan.txt
-rw-rw-r-- 1 arul arul 0 Sep 20 12:37 /tmp/plan.txt
arul@arul-VirtualBox:~/devops_project$ rm /home/arul/devops_project/plan.txt
arul@arul-VirtualBox:~/devops_project$ ls -l /home/arul/devops_project
total 0
-rw-rw-r-- 1 arul arul 0 Sep 20 12:24 network_report.txt
arul@arul-VirtualBox:~/devops_project$ ls -l /tmp/plan.txt
-rw-rw-r-- 1 arul arul 0 Sep 20 12:37 /tmp/plan.txt
arul@arul-VirtualBox:~/devops_project$
```

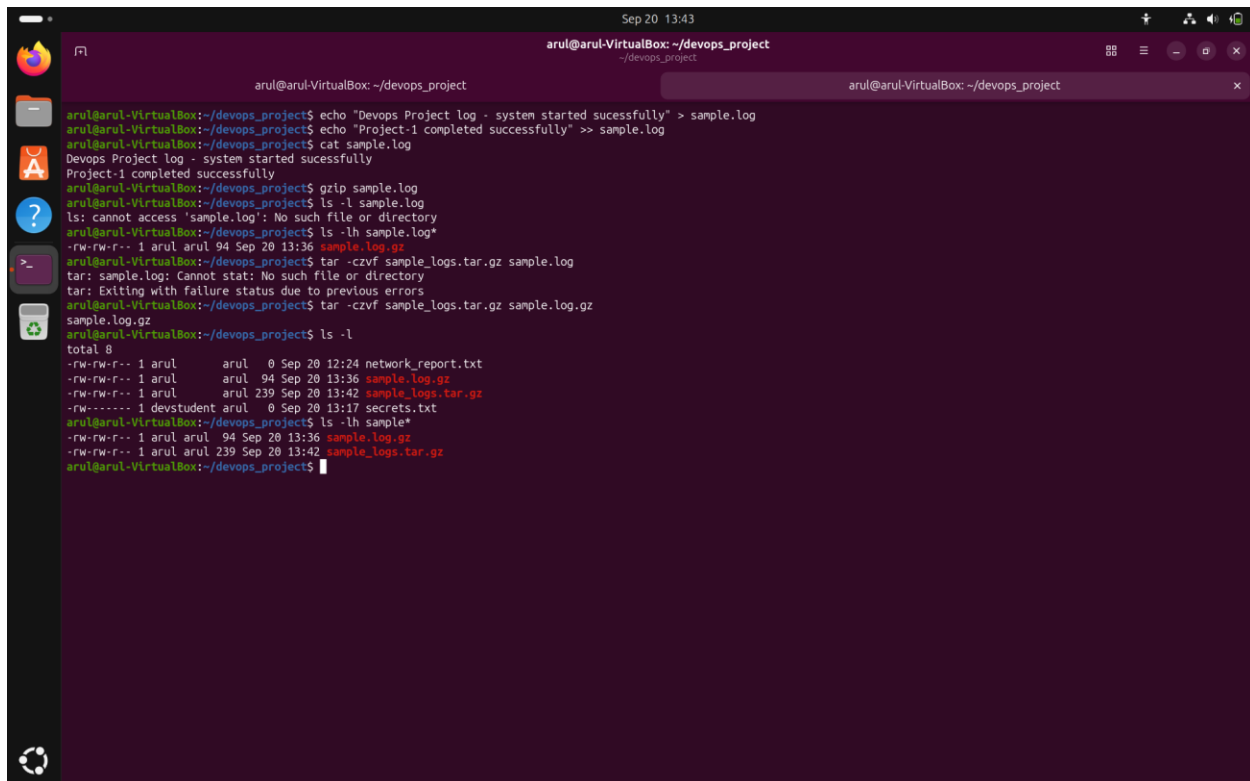
Users & Permissions:

- Create a user devstudent
- Create a group devops_team
- Add devstudent to devops_team
- Create a file secrets.txt, and:
- Change its owner to devstudent
- Set permissions so only owner can read/write

```
Sep 20 13:27
arul@arul-VirtualBox: ~/devops_project
arul@arul-VirtualBox:~/devops_project$ ls
plan.txt
arul@arul-VirtualBox:~/devops_project$ touch network_report.txt
arul@arul-VirtualBox:~/devops_project$ ls
network_report.txt  plan.txt
arul@arul-VirtualBox:~/devops_project$ pwd
/home/arul/devops_project
arul@arul-VirtualBox:~/devops_project$ cp /home/arul/devops_project/plan.txt /tmp/
arul@arul-VirtualBox:~/devops_project$ ls -l /tmp/plan.txt
-rw-rw-r-- 1 arul arul 0 Sep 20 12:37 /tmp/plan.txt
arul@arul-VirtualBox:~/devops_project$ rm /home/arul/devops_project/plan.txt
arul@arul-VirtualBox:~/devops_project$ ls -l /home/arul/devops_project
total 0
-rw-rw-r-- 1 arul arul 0 Sep 20 12:24 network_report.txt
arul@arul-VirtualBox:~/devops_project$ ls -l /tmp/plan.txt
-rw-rw-r-- 1 arul arul 0 Sep 20 12:37 /tmp/plan.txt
arul@arul-VirtualBox:~/devops_project$ sudo adduser devstudent
[sudo: authenticate] Password:
[sudo: authentication failed, try again.]
[sudo: authenticate] Password:
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for devstudent
Enter the new value, or press ENTER for the default
  Full Name []: Arul
   Room Number []:
   Work Phone []:
   Home Phone []:
   Other []:
Is the information correct? [Y/n] y
arul@arul-VirtualBox:~/devops_project$ sudo addgroup devops_team
arul@arul-VirtualBox:~/devops_project$ sudo usermod -aG devops_team devstudent
arul@arul-VirtualBox:~/devops_project$ groups devstudent
devstudent : devstudent users devops_team
arul@arul-VirtualBox:~/devops_project$ touch secrets.txt
arul@arul-VirtualBox:~/devops_project$ ls
network_report.txt  secrets.txt
arul@arul-VirtualBox:~/devops_project$ chown devstudent /home/arul/devops_project/secrets.txt
chown: changing ownership of '/home/arul/devops_project/secrets.txt': Operation not permitted (os error 1)
arul@arul-VirtualBox:~/devops_project$ sudo chown devstudent /home/arul/devops_project/secrets.txt
arul@arul-VirtualBox:~/devops_project$ sudo chmod 600 devstudent /home/arul/devops_project/secrets.txt
chmod: cannot access 'devstudent': No such file or directory
arul@arul-VirtualBox:~/devops_project$ sudo chmod 600 /home/arul/devops_project/secrets.txt
arul@arul-VirtualBox:~/devops_project$ ls -l /home/arul/devops_project/secrets.txt
-rw----- 1 devstudent arul 0 Sep 20 13:17 /home/arul/devops_project/secrets.txt
arul@arul-VirtualBox:~/devops_project$
```

File Compression:

- create dummy one using echo
- Compress it using gzip and tar



The image shows a terminal window titled 'arul@arul-VirtualBox: ~/devops_project' with a dark purple background. The terminal displays the following commands and their outputs:

```
arul@arul-VirtualBox:~/devops_project$ echo "Devops Project log - system started sucessfully" > sample.log
arul@arul-VirtualBox:~/devops_project$ echo "Project-1 completed successfully" >> sample.log
arul@arul-VirtualBox:~/devops_project$ cat sample.log
Devops Project log - system started sucessfully
Project-1 completed successfully
arul@arul-VirtualBox:~/devops_project$ gzip sample.log
arul@arul-VirtualBox:~/devops_project$ ls -l sample.log
ls: cannot access 'sample.log': No such file or directory
arul@arul-VirtualBox:~/devops_project$ ls -lh sample.log*
-rw-rw-r-- 1 arul arul 94 Sep 20 13:36 sample.log.gz
arul@arul-VirtualBox:~/devops_project$ tar -czvf sample_logs.tar.gz sample.log
tar: sample.log: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
arul@arul-VirtualBox:~/devops_project$ tar -czvf sample_logs.tar.gz sample.log.gz
sample.log.gz
arul@arul-VirtualBox:~/devops_project$ ls -l
total 8
-rw-rw-r-- 1 arul arul 0 Sep 20 12:24 network_report.txt
-rw-rw-r-- 1 arul arul 94 Sep 20 13:36 sample.log.gz
-rw-rw-r-- 1 arul arul 239 Sep 20 13:42 sample_logs.tar.gz
-rw-rw-r-- 1 devstudent arul 0 Sep 20 13:17 secrets.txt
arul@arul-VirtualBox:~/devops_project$ ls -lh sample*
-rw-rw-r-- 1 arul arul 94 Sep 20 13:36 sample.log.gz
-rw-rw-r-- 1 arul arul 239 Sep 20 13:42 sample_logs.tar.gz
arul@arul-VirtualBox:~/devops_project$
```

Package Management:

```
Sep 20 13:49
arul@arul-VirtualBox: ~/devops_project

arul@arul-VirtualBox:~/devops_project$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
141 packages can be upgraded. Run 'apt list --upgradable' to see them.
arul@arul-VirtualBox:~/devops_project$ sudo apt install curl
curl is already the newest version (8.18.0-1ubuntu2.5).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 141
arul@arul-VirtualBox:~/devops_project$ sudo apt install htop
Installing:
  htop
Suggested packages:
  ln-sensors
Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 141
  Download size: 177 kB
  Space needed: 451 kB / 13.0 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 htop amd64 3.4.1-5build2 [177 kB]
Fetched 177 kB in 2s (92.2 kB/s)
Selecting previously unselected package htop.
(Reading database... 196146 files and directories currently installed.)
Preparing to unpack .../htop.3.4.1-5build2_amd64.deb...
Unpacking htop (3.4.1-5build2)...
Setting up htop (3.4.1-5build2)...
Processing triggers for desktop-file-utils (0.28-1build1)...
Processing triggers for hicolor-icon-theme (0.18-2build1)...
Processing triggers for gnome-menus (3.38.1-1ubuntu1)...
Processing triggers for man-db (2.13.1-1build1)...
arul@arul-VirtualBox:~/devops_project$ sudo apt install net-tools
net-tools is already the newest version (2.10-2ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 141
arul@arul-VirtualBox:~/devops_project$ curl --version
curl 8.18.0 (x86_64-pc-linux-gnu) libcurl/8.18.0 OpenSSL/3.5.5 zlib/1.3.1 brotli/1.2.0 zstd/1.5.7 libidn2/2.3.8 libpsl/0.21.2 libssh2/1.11.1 nghttp2/1.68.0 librtmp/2.3 mit-krb5/1.22.1 OpenLDAP/2.6.10
Release-Date: 2026-01-07, security patched: 8.18.0-1ubuntu2.5
Protocols: dict file ftp ftps gopher htp https imap imaps ipfs ipns ldap ldaps mqtt pop3 pop3s rtmp rtsp scp sftp smb smbs smtp smtps telnet tftp ws wss
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets zstd
arul@arul-VirtualBox:~/devops_project$ htop --version
htop 3.4.1
```

```
Sep 20 13:50
arul@arul-VirtualBox: ~/devops_project

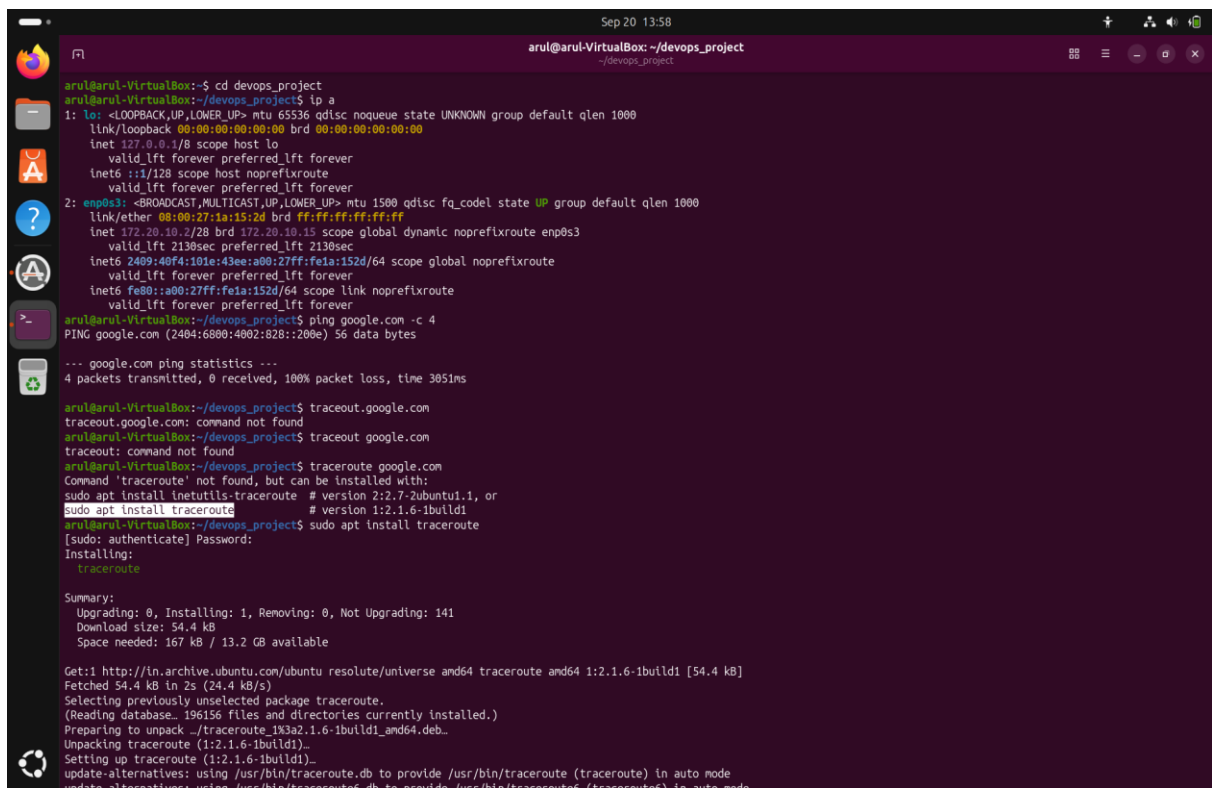
Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 141
  Download size: 177 kB
  Space needed: 451 kB / 13.0 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 htop amd64 3.4.1-5build2 [177 kB]
Fetched 177 kB in 2s (92.2 kB/s)
Selecting previously unselected package htop.
(Reading database... 196146 files and directories currently installed.)
Preparing to unpack .../htop.3.4.1-5build2_amd64.deb...
Unpacking htop (3.4.1-5build2)...
Setting up htop (3.4.1-5build2)...
Processing triggers for desktop-file-utils (0.28-1build1)...
Processing triggers for hicolor-icon-theme (0.18-2build1)...
Processing triggers for gnome-menus (3.38.1-1ubuntu1)...
Processing triggers for man-db (2.13.1-1build1)...
arul@arul-VirtualBox:~/devops_project$ sudo apt install net-tools
net-tools is already the newest version (2.10-2ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 141
arul@arul-VirtualBox:~/devops_project$ curl --version
curl 8.18.0 (x86_64-pc-linux-gnu) libcurl/8.18.0 OpenSSL/3.5.5 zlib/1.3.1 brotli/1.2.0 zstd/1.5.7 libidn2/2.3.8 libpsl/0.21.2 libssh2/1.11.1 nghttp2/1.68.0 librtmp/2.3 mit-krb5/1.22.1 OpenLDAP/2.6.10
Release-Date: 2026-01-07, security patched: 8.18.0-1ubuntu2.5
Protocols: dict file ftp ftps gopher htp https imap imaps ipfs ipns ldap ldaps mqtt pop3 pop3s rtmp rtsp scp sftp smb smbs smtp smtps telnet tftp ws wss
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets zstd
arul@arul-VirtualBox:~/devops_project$ htop --version
htop 3.4.1
arul@arul-VirtualBox:~/devops_project$ ifconfig --version
net-tools 2.10
arul@arul-VirtualBox:~/devops_project$ netstat --version
Command 'netstat' not found, did you mean:
  command 'netrat' from snap netrat (1.0.0)
  command 'netcat' from deb netcat-openbsd (1.234-1)
  command 'netcat' from deb ncat (7.98+dfsg-1)
  command 'netcat' from deb netcat-traditional (1.10-50.1build1)
  command 'netstat' from deb net-tools (2.10-2ubuntu1)
See 'snap info <snapname>' for additional versions.
arul@arul-VirtualBox:~/devops_project$ netstat --version
net-tools 2.10
Fred Baumgarten, Alan Cox, Bernd Eckenfels, Phil Blundell, Tuan Hoang, Brian Micek and others
+NEW_ADDRT +RTF_IRTT +RTF_REJECT +FW_MASQUERADE +I18N +SELINUX
AF: (lnet) +UNIX +INET +INET6 +IPX +AX25 +NETROM +X25 +ATALK +ECONET +ROSE +BLUETOOTH
HM: +ETHER +ARC +SLIP +PPP +TUNNEL -TR +AX25 +NETROM +X25 +FR +ROSE +ASH +SIT +FDDI +HIPPI +HDLC/LAPB +EUI64
arul@arul-VirtualBox:~/devops_project$
```

Networking Commands:

- `ip` - Show the system ip address
 - Network interfaces
 - IPv4 addresses
 - IPv6 addresses
 - MAC addresses
 - Interface status
 - `lo` is the loopback interface.
 - `enp0s3` or similar is usually your VirtualBox network interface.
- `ping google.com -c 4`

`ping` checks whether your machine can communicate with Google's server using ICMP.



```
arul@arul-VirtualBox: ~/devops_project
arul@arul-VirtualBox:~$ cd devops_project
arul@arul-VirtualBox:~/devops_project$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1a:15:2d brd ff:ff:ff:ff:ff:ff
    inet 172.20.10.2/28 brd 172.20.10.15 scope global dynamic noprefixroute enp0s3
        valid_lft 2130sec preferred_lft 2130sec
    inet6 2409:40f4:101e:43ee:a00:27ff:fe1a:152d/64 scope global noprefixroute
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe1a:152d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
arul@arul-VirtualBox:~/devops_project$ ping google.com -c 4
PING google.com (2404:6800:4002:828::200e) 56 data bytes

--- google.com ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3051ms

arul@arul-VirtualBox:~/devops_project$ traceout.google.com
traceout.google.com: command not found
arul@arul-VirtualBox:~/devops_project$ traceout google.com
traceout: command not found
arul@arul-VirtualBox:~/devops_project$ traceroute google.com
Command 'traceroute' not found, but can be installed with:
sudo apt install inetutils-traceroute # version 2:2.7-2ubuntu1.1, or
sudo apt install traceroute # version 1:2.1.6-1build1
arul@arul-VirtualBox:~/devops_project$ sudo apt install traceroute
[sudo: authenticate] Password:
Installing:
  traceroute

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 141
  Download size: 54.4 kB
  Space needed: 167 kB / 13.2 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu/resolute/universe amd64 traceroute amd64 1:2.1.6-1build1 [54.4 kB]
Fetched 54.4 kB in 2s (24.4 kB/s)
Selecting previously unselected package traceroute.
(Reading database... 196156 files and directories currently installed.)
Preparing to unpack ./traceroute_1%3a2.1.6-1build1_amd64.deb...
Unpacking traceroute (1:2.1.6-1build1)...
Setting up traceroute (1:2.1.6-1build1)...
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: using /usr/bin/traceroute6.db to provide /usr/bin/traceroute6 (traceroute6) in auto mode
```

- traceroute google.com
traceroute shows the network hops between your computer and the destination.

- nslookup google.com
Used to query DNS to obtain name or ip address mapping

- netstat -tuln
- -t → TCP connections
- -u → UDP connections
- -l → Listening ports
- -n → Show numerical addresses/ports

```
Sep 20 13:59
arul@arul-VirtualBox: ~/devops_project
update-alternatives: using /usr/sbin/tcptraceroute.db to provide /usr/sbin/tcptraceroute (tcptraceroute) in auto mode
Processing triggers for man-db (2.13.1-1build1)-
arul@arul-VirtualBox:~/devops_project$ traceroute google.com
traceroute to google.com (2404:6800:4002:828::200e), 30 hops max, 80 byte packets
 1  * * *
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * ^Z
[1]+  Stopped                  traceroute google.com
arul@arul-VirtualBox:~/devops_project$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.193.46
Name:   google.com
Address: 2404:6800:4002:828::200e

arul@arul-VirtualBox:~/devops_project$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
tcp6       0      0 :::631                  :::*                     LISTEN
tcp6       0      0 :::22                    :::*                     LISTEN
udp        0      0 0.0.0.0:5353            0.0.0.0:*               LISTEN
udp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN
udp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
udp        0      0 127.0.0.1:323           0.0.0.0:*               LISTEN
udp6       0      0 fe80::a00:27ff:fe1a:546 :::*                     LISTEN
udp6       0      0 :::5353                  :::*                     LISTEN
udp6       0      0 :::1:323                  :::*                     LISTEN
arul@arul-VirtualBox:~/devops_project$
```